

Week 7 — Constraint Qualifications, Envelope Theorem, and Hyperplanes

Review: Mangasarian–Fromovitz Constraint Qualification

Consider the inequality-constrained maximization problem

$$\max_{x \in X} f(x)$$

subject to

$$g_m(x) \leq 0, \quad m = 1, \dots, M,$$

where

$$X \subseteq \mathbb{R}^J$$

is open.

Recall the active set at a feasible point x :

$$\text{Bind}(x) = \{m \in \{1, \dots, M\} : g_m(x) = 0\}.$$

Definition (Mangasarian–Fromovitz constraint qualification). The *Mangasarian–Fromovitz constraint qualification* (MFCQ) holds at a feasible point x^* if there exists a direction

$$v \in \mathbb{R}^J$$

such that

$$\boxed{\nabla g_m(x^*)^\top v < 0 \quad \text{for every } m \in \text{Bind}(x^*)}.$$

Intuition. Since our convention is

$$g_m(x) \leq 0,$$

the gradient $\nabla g_m(x^*)$ points locally toward the side where the constraint becomes larger and therefore infeasible.

Thus

$$\nabla g_m(x^*)^\top v < 0$$

means that moving slightly in direction v moves strictly into the feasible side of the active constraint.

MFCQ requires one single direction v that moves strictly inward with respect to *every* active constraint simultaneously.

Two Simple Ways MFCQ Can Fail

Remark. If an active constraint satisfies

$$\nabla g_m(x^*) = 0,$$

then MFCQ fails.

Indeed, for every $v \in \mathbb{R}^J$,

$$\nabla g_m(x^*)^\top v = 0,$$

so it is impossible to obtain

$$\nabla g_m(x^*)^\top v < 0.$$

Remark. Suppose two constraints g_m and g_n are active at x^* and

$$\nabla g_m(x^*) = -\nabla g_n(x^*).$$

Then MFCQ fails.

Indeed, MFCQ would require simultaneously

$$\nabla g_m(x^*)^\top v < 0$$

and

$$\nabla g_n(x^*)^\top v < 0.$$

But the second inequality is equivalent to

$$-\nabla g_m(x^*)^\top v < 0,$$

or

$$\nabla g_m(x^*)^\top v > 0,$$

which contradicts the first.

Example: Vanishing Gradient of an Active Constraint

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x,$$

subject to

$$g(x) = x^3 \leq 0.$$

Since

$$x^3 \leq 0 \iff x \leq 0,$$

the problem is

$$\max_{x \leq 0} x.$$

Hence the unique solution is

$$x^* = 0.$$

The gradients are

$$f'(x) = 1, \quad g'(x) = 3x^2.$$

At the solution,

$$g'(0) = 0.$$

Therefore there is no direction $v \in \mathbb{R}$ satisfying

$$g'(0)v < 0,$$

so MFCQ fails.

Moreover, the KKT stationarity equation would require

$$1 = \lambda^* \cdot 0,$$

which is impossible.

Remark. This example shows why a constraint qualification is needed in the KKT theorem. The point $x^* = 0$ is genuinely optimal, but no KKT multiplier exists because the active constraint is degenerate at the solution.

Example: Opposite Active Constraint Gradients

Consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x_1, x_2) = x_1,$$

subject to

$$g_1(x_1, x_2) = x_1^2 + x_2 \leq 0$$

and

$$g_2(x_1, x_2) = x_1^2 - x_2 \leq 0.$$

Adding the two inequalities yields

$$2x_1^2 \leq 0,$$

hence

$$x_1 = 0.$$

Then the two constraints imply

$$x_2 \leq 0 \quad \text{and} \quad -x_2 \leq 0,$$

so

$$x_2 = 0.$$

Thus the feasible set contains only

$$x^* = (0, 0),$$

which is therefore the solution.

At x^* ,

$$\nabla g_1(0, 0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \nabla g_2(0, 0) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

Hence

$$\nabla g_1(0, 0) = -\nabla g_2(0, 0),$$

so MFCQ fails.

Moreover,

$$\nabla f(0, 0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

There do not exist $\lambda_1, \lambda_2 \geq 0$ satisfying

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \lambda_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

Thus the KKT conclusion again fails even though x^* is the optimizer.

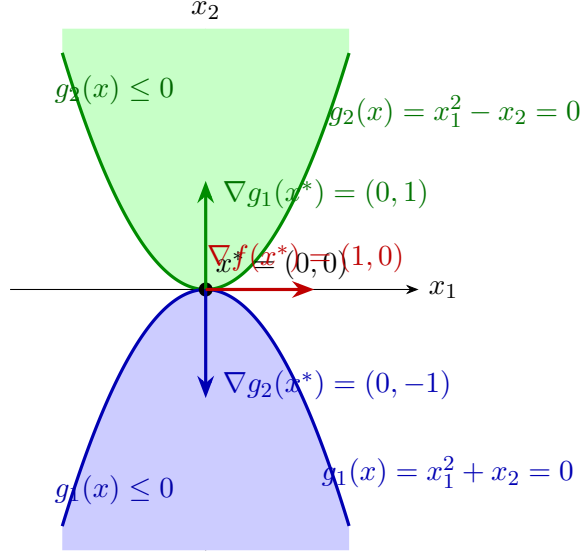


Figure 1: Failure of MFCQ with opposite active constraint gradients. The feasible side of g_1 lies below the lower parabola, while the feasible side of g_2 lies above the upper parabola. Their intersection consists only of $x^* = (0, 0)$. At that point, $\nabla g_1(x^*) = -\nabla g_2(x^*)$, so there is no direction that moves strictly inside both active constraints simultaneously.

The Slater Condition

Definition (Slater condition). The inequality-constrained problem

$$g_m(x) \leq 0, \quad m = 1, \dots, M,$$

satisfies the *Slater condition* if there exists some

$$\bar{x} \in X$$

such that

$$g_m(\bar{x}) < 0 \quad \text{for every } m = 1, \dots, M.$$

Such a point $\bar{x}$ is called a *strictly feasible point*.

Intuition. The Slater condition asks for at least one point at which every inequality constraint has slack.

Geometrically, the feasible set must have a point that lies strictly inside all the inequality constraints rather than being forced entirely onto their boundaries.

Proposition (Slater and MFCQ for convex constraints). *Suppose the feasible set is nonempty, $X \subseteq \mathbb{R}^J$ is open, and each*

$$g_m : X \rightarrow \mathbb{R}$$

is differentiable and convex.

Then the Slater condition holds if and only if MFCQ holds at every feasible point.

Proof. Suppose first that the Slater condition holds. Let $\bar{x}$ be strictly feasible, so

$$g_m(\bar{x}) < 0 \quad \forall m.$$

Fix any feasible x^* . If m is active at x^* , then

$$g_m(x^*) = 0.$$

By convexity,

$$g_m(\bar{x}) \geq g_m(x^*) + \nabla g_m(x^*)^\top (\bar{x} - x^*).$$

Since $g_m(\bar{x}) < 0 = g_m(x^*)$,

$$\nabla g_m(x^*)^\top (\bar{x} - x^*) < 0.$$

Thus the direction

$$v := \bar{x} - x^*$$

satisfies the MFCQ inequalities for every active constraint.

Conversely, suppose MFCQ holds at a feasible point x^* . Then there is a direction v such that

$$\nabla g_m(x^*)^\top v < 0$$

for every active constraint.

For sufficiently small $t > 0$, differentiability implies

$$g_m(x^* + tv) < 0$$

for all active constraints, while inactive constraints remain strictly negative by continuity. Since X is open, $x^* + tv \in X$ for sufficiently small t .

Therefore a strictly feasible point exists, and the Slater condition holds. $\square$

Linear Independence Constraint Qualification

Definition (LICQ). The *Linear Independence Constraint Qualification* (LICQ) holds at a feasible point x^* if the gradients of the active constraints,

$$\{\nabla g_m(x^*) : m \in \text{Bind}(x^*)\},$$

are linearly independent.

Proposition. *LICQ implies MFCQ.*

Proof. Suppose the active gradients are linearly independent. Let

$$A = \begin{pmatrix} \nabla g_{m_1}(x^*)^\top \\ \vdots \\ \nabla g_{m_K}(x^*)^\top \end{pmatrix},$$

where

$$\text{Bind}(x^*) = \{m_1, \dots, m_K\}.$$

LICQ implies

$$\text{rank}(A) = K.$$

Hence the linear system

$$Av = -\mathbf{1}$$

has a solution $v \in \mathbb{R}^J$.

For every active constraint,

$$\nabla g_m(x^*)^\top v = -1 < 0.$$

Thus MFCQ holds. $\square$

Remark. The converse is false:

$$\boxed{\text{LICQ} \implies \text{MFCQ}},$$

but MFCQ does not in general imply LICQ.

Proposition (Uniqueness of KKT multipliers under LICQ). *If LICQ holds at x^* , then the KKT multipliers associated with the active constraints are unique.*

Intuition. Stationarity writes the objective gradient as

$$\nabla f(x^*) = \sum_{m \in \text{Bind}(x^*)} \lambda_m^* \nabla g_m(x^*).$$

If the active gradients are linearly independent, a vector has at most one representation as a linear combination of them. Hence the multipliers are uniquely determined.

Sufficient Conditions for KKT Optimality

The KKT conditions are generally necessary conditions for a local optimum under a suitable constraint qualification. Additional shape restrictions can make them sufficient for global optimality.

Proposition (Sufficiency of KKT). *Let $X \subseteq \mathbb{R}^J$ be open and convex, and suppose*

$$f, g_1, \dots, g_M : X \rightarrow \mathbb{R}$$

are continuously differentiable.

Let x^ be feasible and suppose there exist multipliers $\lambda_m^* \geq 0$ satisfying the KKT conditions.*

Assume that every constraint g_m that binds at x^ is quasiconvex.*

If either

1. f is concave, or

2. f is quasiconcave and

$$\nabla f(x^*) \neq 0,$$

then x^ is a global solution to the inequality-constrained maximization problem.*

Intuition. KKT stationarity says

$$\nabla f(x^*) = \sum_{m \in \text{Bind}(x^*)} \lambda_m^* \nabla g_m(x^*), \quad \lambda_m^* \geq 0.$$

For any feasible alternative y , quasiconvexity of an active constraint implies that its gradient at x^* cannot point toward y :

$$\nabla g_m(x^*)^\top (y - x^*) \leq 0.$$

Therefore

$$\nabla f(x^*)^\top (y - x^*) \leq 0.$$

Under concavity, this first-order inequality rules out any feasible point with higher objective value. Under quasiconcavity and a nonzero gradient, the same geometric argument applies to the convex upper contour set of f .

Interpreting KKT Multipliers

Consider

$$\max_x f(x)$$

subject to

$$g_m(x) \leq 0, \quad m = 1, \dots, M.$$

Suppose the m th constraint binds at the optimum:

$$g_m(x^*) = 0.$$

Now imagine relaxing this constraint slightly. Intuitively:

- If the constraint was slack, a small relaxation should have no first-order effect on the optimum.
- If the constraint was binding, a relaxation enlarges the feasible set and may allow a higher objective value.

The KKT multiplier measures this marginal value of relaxing the constraint.

This interpretation is formalized by the Envelope Theorem.

Parameterized Optimization Problems

Let the choice variable be

$$x \in \mathbb{R}^J$$

and let

$$b \in \mathbb{R}^L$$

be a vector of parameters.

Consider

$$f : \mathbb{R}^J \times \mathbb{R}^L \rightarrow \mathbb{R}$$

and

$$g_m : \mathbb{R}^J \times \mathbb{R}^L \rightarrow \mathbb{R}, \quad m = 1, \dots, M.$$

For each parameter vector b , consider the problem

$$\max_{x \in \mathbb{R}^J} f(x, b)$$

subject to

$$g_m(x, b) \leq 0, \quad m = 1, \dots, M.$$

Suppose there is a solution map

$$h : \mathbb{R}^L \rightarrow \mathbb{R}^J$$

such that

$$x^* = h(b)$$

solves the problem corresponding to parameter b .

Definition (Value function). The *value function* is

$$V : \mathbb{R}^L \rightarrow \mathbb{R}$$

defined by

$$V(b) = f(h(b), b).$$

Thus $V(b)$ records the optimal value of the objective as a function of the parameters.

Define the active set at parameter b by

$$\text{Bind}(b) = \{m : g_m(h(b), b) = 0\}.$$

The Envelope Theorem

Theorem 1 (Envelope theorem). *Suppose that, for parameters near b^* , the optimization problem satisfies the KKT conditions at*

$$x = h(b),$$

with associated multipliers

$$\lambda_1(b), \dots, \lambda_M(b).$$

Assume the solution map and the relevant objects are differentiable at b^ , and let*

$$x^* = h(b^*).$$

Then

$$\nabla V(b^*) = \nabla_b f(x^*, b^*) - \sum_{m \in \text{Bind}(b^*)} \lambda_m(b^*) \nabla_b g_m(x^*, b^*).$$

Since slack constraints have zero KKT multipliers, one may equivalently write

$$\nabla V(b^*) = \nabla_b f(x^*, b^*) - \sum_{m=1}^M \lambda_m(b^*) \nabla_b g_m(x^*, b^*).$$

Why the Indirect Effect Disappears

By definition,

$$V(b) = f(h(b), b).$$

Differentiating gives two types of effects:

1. a *direct effect* of b on the objective;
2. an *indirect effect* through the change in the optimal choice $h(b)$.

By the chain rule,

$$\nabla V(b) = Dh(b)^\top \nabla_x f(h(b), b) + \nabla_b f(h(b), b).$$

At the optimum, KKT stationarity gives

$$\nabla_x f(h(b), b) = \sum_m \lambda_m(b) \nabla_x g_m(h(b), b).$$

For a binding constraint,

$$g_m(h(b), b) = 0.$$

Differentiating this identity yields

$$Dh(b)^\top \nabla_x g_m(h(b), b) + \nabla_b g_m(h(b), b) = 0.$$

Therefore

$$Dh(b)^\top \nabla_x g_m(h(b), b) = -\nabla_b g_m(h(b), b).$$

Substituting into the derivative of the value function eliminates the derivative of the choice rule $Dh(b)$ and yields

$$\nabla V(b) = \nabla_b f(h(b), b) - \sum_m \lambda_m(b) \nabla_b g_m(h(b), b).$$

Intuition. Changing a parameter has two apparent effects:

$$\text{parameter} \longrightarrow \begin{cases} \text{directly changes the objective/constraints,} \\ \text{changes the optimal choice } x^*. \end{cases}$$

The Envelope Theorem says that the second effect does not need to be computed explicitly. Optimality makes the first-order effect through the adjustment of x^* cancel against the first-order constraint adjustments.

This is why we can differentiate the optimized value without explicitly differentiating the optimal policy in the final formula.

Application: Multipliers as Shadow Values

Suppose the original m th constraint is

$$\tilde{g}_m(x) \leq b_m.$$

Rewrite it in standard form as

$$g_m(x, b_m) = \tilde{g}_m(x) - b_m \leq 0.$$

Suppose also that the objective does not directly depend on b_m . Then

$$\frac{\partial f}{\partial b_m} = 0,$$

and

$$\frac{\partial g_m}{\partial b_m} = -1.$$

The Envelope Theorem therefore gives

$$\frac{\partial V}{\partial b_m} = -\lambda_m^* \frac{\partial g_m}{\partial b_m} = -\lambda_m^* (-1).$$

Hence

$$\boxed{\frac{\partial V}{\partial b_m} = \lambda_m^*}.$$

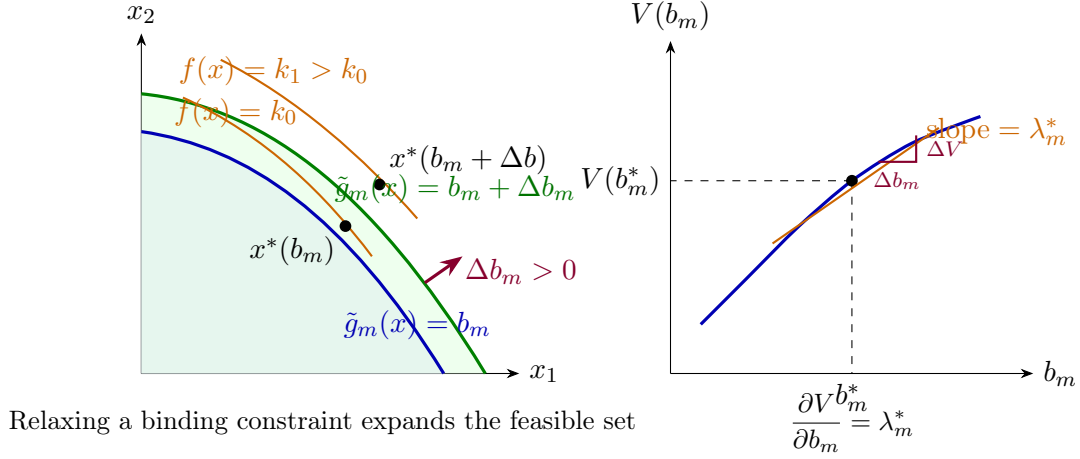


Figure 2: Shadow-price interpretation of the KKT multiplier. Relaxing a binding constraint from $\tilde{g}_m(x) \leq b_m$ to $\tilde{g}_m(x) \leq b_m + \Delta b_m$ expands the feasible set and may raise the optimal value. The Envelope Theorem implies that the marginal value of this relaxation is $\partial V / \partial b_m = \lambda_m^*$.

Intuition (Shadow price). The multiplier λ_m^* is the marginal value of relaxing the m th resource constraint.

A small increase

$$db_m > 0$$

changes the optimal value approximately by

$$dV \approx \lambda_m^* db_m.$$

Thus λ_m^* is often called the *shadow price* of the constraint.

Remark. If the constraint is slack at the optimum, complementary slackness implies

$$\lambda_m^* = 0.$$

Therefore

$$\frac{\partial V}{\partial b_m} = 0.$$

This matches the intuition: slightly relaxing a constraint that was not binding does not change the optimum to first order.

Separating Hyperplanes

Hyperplanes

A hyperplane in $\mathbb{R}^J$ is the higher-dimensional analogue of a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$.

It is an affine set of dimension $J - 1$.

- In $\mathbb{R}$, a hyperplane is a point.
- In $\mathbb{R}^2$, a hyperplane is a line.

- In $\mathbb{R}^3$, a hyperplane is a plane.

Definition (Hyperplane). Let

$$p \in \mathbb{R}^J \setminus \{0\}, \quad \alpha \in \mathbb{R}.$$

The hyperplane generated by (p, α) is

$$H[p, \alpha] = \{x \in \mathbb{R}^J : p^\top x = \alpha\}.$$

Remark. A hyperplane is generally an *affine* set rather than a vector subspace.

It is a vector subspace only in the special case

$$\alpha = 0,$$

because only then does it necessarily contain the origin.

Low-Dimensional Cases

If $J = 1$, then

$$px = \alpha, \quad p \neq 0,$$

so

$$H[p, \alpha] = \left\{ \frac{\alpha}{p} \right\}.$$

Thus a hyperplane in $\mathbb{R}$ is a single point.

If $J = 2$, then

$$H[p, \alpha] = \{(x_1, x_2) \in \mathbb{R}^2 : p_1 x_1 + p_2 x_2 = \alpha\}.$$

Take two points

$$x = (x_1, x_2), \quad y = (y_1, y_2)$$

in the hyperplane. Then

$$p_1 x_1 + p_2 x_2 = p_1 y_1 + p_2 y_2.$$

Therefore

$$p_1(x_1 - y_1) = -p_2(x_2 - y_2).$$

If $p_2 \neq 0$ and $x_1 \neq y_1$,

$$\frac{x_2 - y_2}{x_1 - y_1} = -\frac{p_1}{p_2}.$$

Thus the hyperplane has constant slope.

Why the Normal Vector Must Be Nonzero

Suppose instead that

$$p = 0.$$

Then

$$H[0, \alpha] = \{x \in \mathbb{R}^J : 0 = \alpha\}.$$

Therefore

$$H[0, \alpha] = \begin{cases} \emptyset, & \alpha \neq 0, \\ \mathbb{R}^J, & \alpha = 0. \end{cases}$$

Neither object is the codimension-one affine set we mean by a hyperplane. This is why the definition requires

$$p \neq 0.$$

Equivalent Representations of a Hyperplane

For every scalar

$$\lambda \neq 0,$$

we have

$$\boxed{H[p, \alpha] = H[\lambda p, \lambda \alpha].}$$

Indeed,

$$p^\top x = \alpha \iff (\lambda p)^\top x = \lambda \alpha.$$

Thus the pair (p, α) representing a hyperplane is not unique.

We may normalize the normal vector. Define

$$q = \frac{p}{\|p\|}, \quad \beta = \frac{\alpha}{\|p\|}.$$

Then

$$\|q\| = 1$$

and

$$H[p, \alpha] = H[q, \beta].$$

The Normal Vector

Fix

$$y \in H[p, \alpha].$$

Since

$$p^\top y = \alpha,$$

we have

$$x \in H[p, \alpha]$$

if and only if

$$p^\top x = p^\top y.$$

Equivalently,

$$\boxed{p^\top (x - y) = 0.}$$

Thus every displacement vector $x - y$ lying along the hyperplane is orthogonal to p .

Therefore p is called a *normal vector* to the hyperplane.

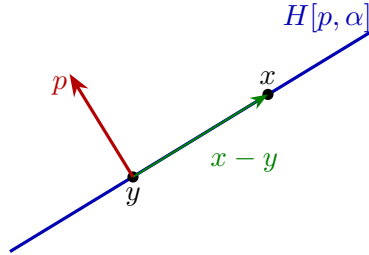


Figure 3: The normal vector p is perpendicular to every direction lying inside the hyperplane.

Halfspaces

A hyperplane divides $\mathbb{R}^J$ into two sides.

Definition (Closed halfspaces). The two closed halfspaces determined by $H[p, \alpha]$ are

$$H^+[p, \alpha] = \{x \in \mathbb{R}^J : p^\top x \geq \alpha\}$$

and

$$H^-[p, \alpha] = \{x \in \mathbb{R}^J : p^\top x \leq \alpha\}.$$

Their common boundary is

$$H[p, \alpha].$$

Definition (Open halfspaces). The corresponding open halfspaces are

$$\{x \in \mathbb{R}^J : p^\top x > \alpha\}$$

and

$$\{x \in \mathbb{R}^J : p^\top x < \alpha\}.$$

Separating Sets by a Hyperplane

The basic geometric idea is to place two sets on opposite sides of a hyperplane.

Definition (Separating hyperplane). Let

$$A, B \subseteq \mathbb{R}^J.$$

We say that the hyperplane

$$H[p, \alpha]$$

separates A and B if either

$$A \subseteq \{x : p^\top x \geq \alpha\}$$

and

$$B \subseteq \{x : p^\top x \leq \alpha\},$$

or

$$A \subseteq \{x : p^\top x \leq \alpha\}$$

and

$$B \subseteq \{x : p^\top x \geq \alpha\}.$$

Definition (Separated sets). We say that A and B are *separated* if there exist

$$p \in \mathbb{R}^J \setminus \{0\}$$

and

$$\alpha \in \mathbb{R}$$

such that $H[p, \alpha]$ separates A and B .

Remark. This is a *weak* notion of separation because the sets are allowed to touch the separating hyperplane.

Thus separation does not necessarily imply

$$A \cap B = \emptyset.$$

However, if a point belongs to both A and B , then under the separation conditions it must lie on the hyperplane:

$$A \cap B \subseteq H[p, \alpha].$$

Intuition. A separating hyperplane creates a linear boundary such that one set lies entirely on one side and the other set lies entirely on the opposite side.

The importance of convexity will become clear in the separating hyperplane theorems: convex sets have enough geometric structure for linear boundaries to describe their relative position.

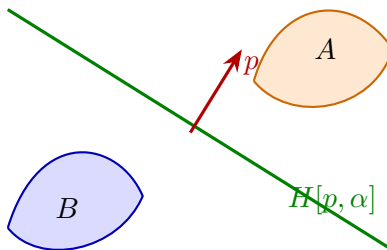


Figure 4: A hyperplane separates two sets by placing them in opposite halfspaces.